Mini-Challenge DEL: Hyperparameter Tuning and Model Evaluation

Goal

Implement, train, and validate several neural network models for a classification task using a suitable dataset. In the process, get to know some techniques of deep learning.

Procedure

The task to be modeled and the appropriate data are to be agreed upon individually at the beginning of the mini-challenge. You can also bring in your own ideas and propose a task and dataset.
You will create a Jupyter Notebook, which you will submit in two stages:

1. **First stage**: Submission with the baseline model and evaluated learning rate and batch size (after "Step 2" below);
2. **Second stage**: Final submission after all steps.

First Stage: Baseline Model

Choosing Task / Dataset

1. Consider what task should be learned and which data you would like to work with.
   Note: When selecting the dataset, care must be taken that the focus of the mini-challenge can remain on training and evaluation, without spending too much time on data acquisition, analysis, or preprocessing. The dataset should also not be too large, so that many different model and parameter variants can be evaluated.
2. Specify a proposal for a model that you would like to use as your baseline.
   Note: The baseline model should be an architecture with a few convolutional layers for feature extraction and a few fully connected layers for classification. From this starting point, various variants will be examined (see below).

Agreement with the coach about data, task, and baseline model.

Data Familiarization, Model Building, Baseline Model

1. Data preparation:
   - (Short) exploratory analysis of the features (e.g. comparison of classes per feature, class balance).
   - Suitable preprocessing (e.g. data normalization).
2. Define the metrics for evaluation.
3. Implement basic functionality:
   - For training and evaluating models.
   - Use an MLOps platform (e.g. W&B).
   - Implement the agreed baseline model.
   - Overfitting test: Train the baseline model with a single sample or one batch to test the basic functionality.
4. Training the baseline model:
   - Train with SGD (without momentum), without regularization, without batch normalization, using all training data.

- Tune the learning rate and batch size. Show differences using the learning curves with the chosen loss function and metrics. Discuss the results and compare them with theoretical expectations.
  Note: Do not use automatic hyperparameter search methods (e.g. no Bayesian optimization).
- Estimate the statistical error of these metrics (using cross-validation, performed at least for one training run).

Coordination with the coach:

- **Submission (first stage):**
  - Show the notebook to the coach by walking through it.
  - Share the notebook with the coach (frozen git version).
- **Discussion with the coach:**
  - The coach gives feedback on the first submission, with which you should create an improved version.
  - Proceed with the second stage.

Second Stage: Hyperparameter Tuning

Next, you should investigate the training process and model performance when changing various hyperparameters (see below). For this you should formulate at least **10 hypotheses**, conduct experiments to test them, and analyze and reflect on the results.

For each hypothesis, proceed as follows:

a) *Hypothesis*: Formulate your expectation about what will change in model performance and/or training behavior when a hyperparameter is changed. Try to justify this hypothesis using your theoretical knowledge (to be developed if necessary). Provide an appropriate reference.
b) *Experiment*: Define an experiment to test the hypothesis. Conduct the experiment.
c) *Analysis/Reflection:* Analyze whether the results confirm the hypothesis. Reflect on them.

The hypotheses should come from the following topic groups:

- Model complexity:
  - Number of convolutional layers and/or number of linear layers (depth).
  - Number of filters per convolutional layer or number of units in linear layers (width).
  - (You may need to adjust the learning rate again.)
- Settings in convolutional layers:
  - Kernel size
  - Pooling
- Regularization:
  - Dropout
  - Data augmentation
- Use of BatchNorm
- Weight initialization
- Use of Adam optimizer
- Transfer learning

Finally, you should draw conclusions about which variants/methods contribute to a better-functioning model.

Then, build *"your best model"* from a suitable combination of the above methods.

Agreement with the coach: Show the coach the first formulated hypothesis and the corresponding experiment. Discuss further hypotheses with the coach as needed.

Presentation, Report

1. Presentation (~10 minutes): Short presentation (slides, not notebook) with discussion of the most important results.
2. Q&A (~10 minutes): Questions to test understanding of the examined methods (SGD, parameter tuning, regularization, batch normalization, optimizers, transfer learning).
3. Report in the form of one (!) well-documented, clearly structured Jupyter Notebook (typically the notebook from the first stage, expanded). This should allow the coach to follow the steps. Outputs (e.g. from W&B) should be integrated in the notebook, so that the coach does not have to rerun all the trainings.

Timeline

To be agreed with coach at beginning.

Assessment

Assessment is based on the submitted notebook and the presentation incl. Q&A:

- Complete and correct implementation of the agreed assignment.
- Clear, well-structured execution.
- Coherent and conzise description and interpretation of results. Well-chosen and well-commented plots and tables. Verbose, by ChatGPT generated text will negatively contribute to the grade.
- Sensible use of (computing) resources.
- Understandable presentation of results, correct answers to questions.